IDENTIFYING TAKOTSUBO SYNDROME IN CT SCANS

A thesis submitted to The University of Manchester
for the degree of Doctor of Philosophy
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Abstract

Identifying Takotsubo Syndrome in CT Scans
Awab Akram
A thesis submitted to The University of Manchester
for the degree of Doctor of Philosophy, 2025

This thesis answers all the questions you were afraid to ask ...

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I would like to thank...

Chapter 1

Introduction

1.1 Project Motivation

Takotsubo syndrome, also known as "broken heart syndrome," is a temporary heart condition that often mimics the symptoms of a heart attack. Due to this similarity, it is frequently misdiagnosed, leading to inappropriate treatments and potentially adverse outcomes for patients. The condition is characterized by a temporary weakening of the left ventricle, the heart's main pumping chamber, which changes shape to resemble a Japanese octopus trap (takotsubo) – hence the name [1].

The accurate diagnosis of Takotsubo syndrome typically requires a combination of clinical evaluation, electrocardiography, biomarker testing, and advanced imaging techniques. However, the subtle presentation of this syndrome in imaging makes it challenging to identify, particularly in the early stages when it most closely resembles a myocardial infarction.

This project is motivated by the need to improve the diagnostic accuracy of Takotsubo syndrome through the application of machine learning techniques to analyze CT scans. By developing a model capable of identifying the characteristic features of Takotsubo syndrome in imaging data, we aim to facilitate earlier and more accurate diagnosis, ultimately improving patient outcomes.

1.2 Project Objectives

The primary objective of this project is to develop a machine learning model that can accurately identify Takotsubo syndrome in CT scans. Specifically, the project aims to:

1. Process and prepare DICOM files from both normal and Takotsubo cases for analysis.

2. Implement effective segmentation techniques to isolate the heart region in CT scans.

3. Develop and train a 3D Convolutional Neural Network (CNN) to classify CT scans.
4. Evaluate the model's performance using appropriate metrics.
5. Create an interpretable model that can assist clinicians in diagnosis.

By achieving these objectives, the project seeks to create a tool that can serve as a decision support system for clinicians, flagging potential cases of Takotsubo syndrome for further investigation and appropriate management.

1.3 Report Structure

The remainder of this report is structured as follows:

- **Background**: Provides a comprehensive overview of Takotsubo syndrome, its clinical presentation, diagnostic challenges, and current approaches. It also introduces machine learning concepts relevant to medical imaging analysis.
- **Design**: Details the methodology, including data preparation, model architecture, and implementation strategies.
- **Results**: Presents the performance of the model, including classification metrics and validation results.
- **Conclusion**: Summarizes the findings, discusses limitations, and suggests directions for future work.

Chapter 2

Background

2.1 Takotsubo Syndrome (TTS) Overview

2.1.1 Definition

Takotsubo syndrome (TTS), also known as Takotsubo Cardiomyopathy or "Broken Heart Syndrome," is an acute but reversible form of heart failure that is often triggered by emotional or physical stress [1]. The condition is characterized by transient left ventricular dysfunction in the absence of obstructive coronary artery disease. Internally, TTS presents with a distinct pattern of left ventricular wall motion abnormality, most notably apical ballooning, where the apex of the heart dilates while the base remains hypercontractile. This results in the ventricle adopting a shape reminiscent of a traditional Japanese octopus trap, inspiring the name "Takotsubo" [2].

Externally, patients may present with symptoms such as chest pain, shortness of breath, nausea, and fatigue [3]. These symptoms closely resemble those of acute myocardial infarction (AMI), making initial diagnosis challenging [4, 5]. Unlike AMI, however, TTS is non-ischemic—meaning it is not caused by blocked coronary arteries—though some patients may have coexisting coronary artery disease [4, 6]. The underlying pathophysiology is not yet fully understood, but it is widely believed to involve a surge in catecholamines (stress hormones), resulting in myocardial stunning through mechanisms such as microvascular spasm, energy imbalance, or direct toxicity [7].

2.1.2 Epidemiology

TTS is relatively rare but likely underdiagnosed. It is estimated to account for approximately 1–3% of all cases of suspected ST-Elevation Myocardial Infarction (STEMI) [8]. The condition disproportionately affects postmenopausal women, with up to 90% of cases occurring in this demographic and a mean age of around 67 years [8]. Women over the age of 55 are particularly at risk, with studies suggesting they may be up to five times

more likely to develop TTS than younger women or men [8]. This gender imbalance is thought to be influenced by hormonal changes, particularly reduced estrogen levels following menopause.

Although rare, TTS can also occur in younger individuals and even in children, though such cases are uncommon [9]. Increased clinical awareness in recent years has led to more frequent recognition of the syndrome, including among male patients, particularly when physical stressors are involved. Recurrence is possible, with estimates suggesting a rate of approximately 1.8% per patient-year [10], highlighting the importance of long-term follow-up and patient monitoring.

2.1.3 Why is Takotsubo Syndrome Frequently Misdiagnosed?

Takotsubo syndrome (TS) is often misdiagnosed as acute myocardial infarction (AMI) due to several factors:

1. **Similar clinical presentation:** TTS and AMI present with nearly identical symptoms—chest pain, dyspnea, and syncope — making initial differentiation difficult [8].

2. **ECG and Biomarker Overlap:** TTS commonly shows ST-segment elevations and T-wave inversions, similar to AMI [10]. Cardiac troponins may be modestly elevated, further contributing to confusion [2].

3. **Limited access to immediate advanced imaging:** Definitive diagnosis of TTS often requires advanced imaging techniques like echocardiography or cardiac MRI to identify the characteristic wall motion abnormalities. In emergency settings, these imaging modalities may not be immediately available, leading to initial misdiagnosis based on more readily accessible tests [11].

4. **Coexisting Coronary Artery Disease (CAD):** Some TTS patients may have underlying coronary artery disease, which can confound the diagnosis and lead clinicians to attribute the symptoms to AMI [10].

5. **Lack of awareness:** Despite increasing recognition, many clinicians may still be unfamiliar with TTS, especially in regions where it is less commonly encountered [12].

6. **Evolving diagnostic criteria:** The diagnostic criteria for TTS have evolved over time, and there is still no universal consensus, which can contribute to misdiagnosis [8].

7. **Atypical variants:** While the apical variant is most common, TTS can present with other patterns of left ventricular dysfunction, such as midventricular or basal

variants. These atypical presentations may be more challenging to recognize and can be mistaken for other cardiac conditions [8].

2.1.4 Diagnostic Criteria and Guidelines

Diagnosing Takotsubo Syndrome (TTS) typically follows one of two established frameworks: the **Mayo Clinic Criteria** or the **InterTAK Diagnostic Criteria**, supported by the European Society of Cardiology (ESC).

Mayo Clinic Criteria

The Mayo Clinic Criteria, revised in 2008, require all four of the following to be met:

1. Transient left ventricular wall motion abnormalities [4]
2. Absence of obstructive coronary artery disease on angiography [4]
3. New ECG changes (such as ST elevation or T-wave inversion) or elevated cardiac troponin [4]
4. No evidence of pheochromocytoma or myocarditis [4]

Although these criteria originally excluded patients with any coronary artery disease (CAD), it is now accepted that **TTS can occur alongside mild or unrelated CAD**, as long as it doesn't explain the wall motion abnormality[10].

ESC and InterTAK Criteria

The ESC and InterTAK guidelines build on the Mayo framework but allow for more flexibility. They accept that TTS can occur with incidental CAD and include neurological triggers like seizures and stroke as possible causes [8]. **The diagnosis remains one of exclusion**, confirmed after ruling out heart attacks and myocarditis, usually using a combination of imaging and lab results.

2.1.5 Consequences of Misdiagnosis

While TTS is often reversible, misdiagnosis can lead to inappropriate treatments that may cause further complications. Since its symptoms resemble those of a heart attack, patients with TTS are frequently subjected to therapies designed for ischemic heart disease, such as anti-thrombotic medications or invasive procedures[1, 2].

Additionally, TTS can lead to serious complications such as heart failure, arrhythmias, left ventricular outflow tract obstruction, or even cardiogenic shock in severe cases [1]. Early and accurate diagnosis is essential to ensuring proper management and reducing the risk of adverse outcomes.

2.2 Diagnostic Imaging for Takotsubo Syndrome

2.2.1 Heart Tracing (ECG)

An electrocardiogram (ECG) records the electrical activity of the heart. In Takotsubo syndrome (TTS), ECG often exhibits abnormalities that mimic acute myocardial infarction, such as ST-segment elevation and T-wave inversion. Additionally, QT interval prolongation is commonly observed [9]. However, these changes are typically less pronounced than those seen in true myocardial infarctions and lack reciprocal changes, which are indicative of transmural ischemia [4].

2.2.2 Blood Tests

Cardiac biomarkers can help differentiate TTS from myocardial infarction. Troponin levels, which indicate myocardial injury, are only mildly elevated in TTS, whereas they are significantly higher in myocardial infarction [9]. Conversely, B-type natriuretic peptide (BNP) or its precursor NT-proBNP is usually substantially elevated in TTS due to myocardial stretching without permanent damage [1].

2.2.3 Heart Imaging

Echocardiography

This ultrasound-based technique reveals distinctive patterns of left ventricular wall motion abnormalities in TTS, most commonly apical ballooning, where the apex is akinetic while the base contracts normally [13]. These abnormalities usually extend beyond the distribution of a single coronary artery.

Coronary Angiography and CT

In TTS, coronary angiography or CT angiography typically shows unobstructed or only mildly narrowed coronary arteries, helping to distinguish TTS from acute coronary syndromes caused by occlusive coronary artery disease [7].

Cardiac MRI

Cardiac magnetic resonance imaging can demonstrate myocardial edema and hyperemia in TTS but usually lacks the late gadolinium enhancement indicative of necrosis, thus helping to exclude myocardial infarction [14].

2.2.4 Comparison of Diagnostic Modalities

Table 2.1: Comparison of Diagnostic Modalities in TTS vs. Acute Myocardial Infarction (AMI)

Modality	**Typical Findings in TTS**	**Distinction from AMI**
ECG	ST-segment elevation, T-wave inversion, prolonged QT interval [9]	Changes are less pronounced; lacks reciprocal changes [4]
Blood Tests	Mildly elevated troponin [9]; significantly elevated BNP/NT-proBNP [1]	Troponin is much more elevated [9]; BNP is usually lower [1]
Echocardiography	Apical ballooning; apex akinetic, base contracts normally [13]	Wall motion abnormalities extend beyond one artery's territory [4]
Coronary Angiography / CT	Normal or mildly narrowed coronary arteries [7]	AMI typically involves occlusion of major arteries [4]
Cardiac MRI	Myocardial edema and hyperemia without late gadolinium enhancement [14]	AMI shows necrosis with late gadolinium enhancement [14]

2.3 Machine Learning in Medical Imaging

Recent years have seen an explosion in the application of **machine learning (ML)** techniques to medical data, particularly imaging. Machine learning refers to a set of computational methods that enable computers to **learn patterns from data** and improve their performance on tasks without being explicitly programmed for each rule. A subset of ML, **deep learning (DL)**, uses multi-layered neural networks to automatically learn hierarchical representations of data. This section provides an introduction to machine learning and deep learning in the context of medical imaging, focusing on convolutional neural networks (CNNs) which are the workhorse for image analysis tasks. We also explain the difference between 2D and 3D CNN approaches and their relevance to analyzing volumetric scans such as CT images.

2.3.1 What is Machine Learning?

In traditional programming, a human defines explicit rules to process data and produce an output. In machine learning, by contrast, the algorithm **learns the rules or patterns from examples**. In supervised ML (common in medical imaging), we provide the model with input data (e.g., an image) and a desired output (e.g., diagnosis), and the

model iteratively adjusts internal parameters to map the input to the correct output. If we have enough examples (training data), the model uncovers statistical patterns that allow it to make predictions on new, unseen data. For instance, an ML model could be trained on many labeled cardiac images (some with TTS, some with AMI) and learn to distinguish them. A classic ML approach might involve manually designing features (like measuring wall motion or shape descriptors) and then using a classifier (like logistic regression or a decision tree) to combine those features. However, manual feature design is challenging for complex images. This is where deep learning revolutionized the field – **deep neural networks can automatically learn features** directly from raw images, often surpassing human-crafted features in efficacy [15].

In medical imaging, ML has been applied to tasks such as detecting tumors in scans, classifying lesions, segmenting organs, etc., with impressive results [16]. Notably, around 2012, deep learning began outperforming previous methods in image recognition, and since then it has become the dominant approach for image analysis [17].

2.3.2 Neural Networks and Deep Learning

Neural networks are computational models inspired by the structure of the brain, consisting of layers of interconnected artificial neurons. Each neuron computes a weighted sum of its inputs and passes it through a nonlinear activation function. In a **deep neural network**, multiple hidden layers allow the model to learn complex hierarchical representations. Deep learning has enabled automatic feature extraction from raw data, eliminating the need for handcrafted features [18].

In medical imaging, deep learning models—particularly convolutional neural networks—have demonstrated state-of-the-art performance across tasks like detection, classification, segmentation, and registration [19]. The success of deep learning is largely attributed to the availability of large annotated datasets, powerful GPUs, and open-source frameworks such as **PyTorch**, which provide efficient tools for model development and training [20]. PyTorch offers dynamic computation graphs, GPU acceleration, and modular APIs, making it a widely used library in the medical imaging research community.

2.3.3 Convolutional Neural Networks (CNNs)

A convolutional neural network is a type of deep learning model specifically designed for grid-like data such as images. In a regular neural network, each input (pixel) connects to neurons in the next layer (fully-connected), which doesn't scale well to large images. CNNs introduce the concept of **convolutions** – sliding filters (kernels) across the image to detect local patterns. A CNN typically consists of multiple layers: convolutional layers that apply filters to extract features (like edges, textures, shapes), pooling layers that downsample or summarize regions (for spatial invariance and to reduce computation),

and eventually fully-connected layers that make a classification decision based on the extracted features.

The convolution operation is key: a small filter (for example 3×3 pixels) is scanned over the image and computes a weighted sum at each position, producing a feature map that highlights where certain features are present. Early layers might detect simple features (edges or blobs), while deeper layers detect complex concepts (like a ventricular shape). The CNN learns the filter weights during training.

2D CNNs operate on 2D images (height × width), which is ideal for single slice images like chest X-rays or pathology slides. **3D CNNs** operate on 3D data (height × width × depth), which is suitable for volumetric scans like CT or MRI where an additional dimension (slices or time) provides context [21].

CNNs have become the dominant architecture for medical image analysis because of their ability to automatically learn relevant visual features. For example, if we wanted a CNN to identify Takotsubo Syndrome on an imaging exam, we would feed in a set of scans labeled as TTS or not TTS. The CNN's convolutional filters in early layers might learn to detect shapes like the ballooned apex or ventricular contours, mid-level layers might combine those to detect a pattern of wall motion abnormality, and the final layers would learn to associate those patterns with the diagnosis of TTS. All this is learned from data, rather than explicitly programmed. This ability to learn subtle and complex patterns is crucial in medicine, where differences can be nuanced. Indeed, deep learning algorithms, in particular CNNs, have rapidly become a methodology of choice for analyzing medical images [16].

2.3.4 2D vs. 3D CNNs

The distinction between 2D and 3D CNNs is important for medical imaging, because many medical images are inherently three-dimensional (like a stack of CT slices). A **2D CNN** treats input data as flat images. If applied to volumetric data, a 2D CNN would typically either analyze single slices independently or combine results from multiple 2D views. In contrast, a **3D CNN** can take a volume (a stack of slices) as input and has filters that extend in all three dimensions (x, y, and z). This means a 3D CNN can learn features that depend on the relationship between slices (the depth dimension). For example, in a cardiac CT, a 3D convolution could capture an apical ballooning pattern that spans across several adjacent slices of the heart. A 2D CNN might have to infer such a 3D pattern indirectly or require multiple 2D views.

There are trade-offs: 3D CNNs use more memory and computational power because of the extra dimension in filters and data. They also need more training data to learn the additional parameters. However, they can potentially capture the full spatial context of the anatomy. In tasks like detecting tumors in a CT or analyzing heart function

from MRI, 3D CNNs have shown superior accuracy because they evaluate the volumetric context [21]. On the other hand, 2D CNNs can leverage pretrained models from broad image databases (like ImageNet) and may be easier to train with limited data. Sometimes a hybrid approach called "2.5D" is used: for instance, feeding multiple adjacent slices into a 2D CNN (as separate channels) to give it some 3D context without full 3D convolution.

For the task of detecting Takotsubo Syndrome in CT scans (which are 3D by nature), a 3D CNN approach could analyze the entire heart volume in one go, potentially learning the three-dimensional shape deformation of the ventricle in TTS. A 2D CNN could be used by analyzing multiple 2D views (like apical 4-chamber, short-axis slices, etc.), but may not capture global shape as directly. Researchers often experiment with both: a 2D CNN might be simpler and need fewer data, while a 3D CNN might capture the telltale features of TTS more holistically if sufficient training data is available [19].

2.4 Image Segmentation in Cardiac Imaging

2.4.1 Definition

Image segmentation is the process of partitioning an image into meaningful regions. In medical imaging, segmentation typically means delineating anatomical structures or abnormalities – for example, outlining the heart chambers on a CT scan, or segmenting a tumor from healthy tissue. The goal is to assign a label to each pixel or voxel (e.g., "this voxel is part of the left ventricle, this voxel is background"). Segmentation is fundamental because it transforms raw images into quantitative data (such as volumes, shapes, and locations of structures) that can be analyzed or used by diagnostic algorithms. In cardiac imaging, segmentation can isolate the myocardium, ventricles, atria, or even the coronary vessels, enabling measurements of heart function (ejection fraction, wall thickness) and detection of regional motion abnormalities [22, 23, 24].

2.4.2 Importance in Cardiac CT

In the context of cardiac CT scans, segmentation is often a crucial preprocessing step. For instance, to detect TTS features, one might first segment the left ventricle from the CT to focus the analysis on the myocardium. Segmentation allows extraction of metrics like left ventricular volume, which could show an akinetic apex (if segmental volumes or motion are calculated). It also enables visualization tools (like overlaying strain or heatmaps on the heart). Furthermore, if one wants to apply machine learning, having segmented structures can reduce the complexity of the task – e.g., the algorithm can analyze just the heart region instead of the whole chest CT. Segmentation is also used for planning treatments (like identifying myocardial scar vs healthy tissue, though that's more in MRI)

and for research (such as measuring the curvature of the LV apex in TTS) [22, 25].

Manual segmentation by expert radiologists or cardiologists is considered the gold standard for accuracy, but it is **time-consuming and labor-intensive** [26, 27]. Given the complex shape of the heart and possibly low contrast between blood and myocardium on non-contrast scans, manual outlining is tedious. Thus, automated segmentation methods are highly desirable.

2.4.3 Segmentation Methods

Over the years, various approaches have been developed:

- **Threshold and Region-Based Methods:** Early techniques segment images by intensity thresholds or region growing. For example, in CT, setting a high threshold can segment bones easily (since bone is very bright). For the heart, thresholding might separate blood pool from myocardium if contrast is used. However, classical segmentation methods (thresholding, edge detection, region growing) often struggle with cardiac CT because of **noise and low contrast** differences between myocardium and surrounding tissues. The heart borders may not be sharply defined and can be confounded by other structures (lungs, liver). These methods lack adaptability – a global threshold might not work across an entire image if lighting varies or if pathologies alter intensities [28, 29].

  text

- **Contour-Based (Active Contours and Level Sets):** These techniques introduce a deformable model or "snake" that moves under algorithmic guidance to lock onto edges of structures. Active contour models have been widely used in heart segmentation due to their ability to produce smooth, continuous boundaries. For example, one can initialize a contour roughly around the left ventricle and an energy minimization process pulls the contour towards strong edges (the endocardial border). While active contours can be accurate and **provide continuous boundaries**, they are **sensitive to initialization and noise**. If the initial guess is far off or the image edges are weak, the contour might converge to an incorrect boundary or require heavy tuning of parameters [30, 31].

- **Statistical Shape Models:** Another approach uses predefined shape models of the heart (perhaps built from annotated datasets) and fits them to the new image. The model knows the typical geometry of a left ventricle, for example, and an algorithm deforms this model to match the image data. This incorporates prior anatomical knowledge and can deal with missing or noisy edges. Shape models have shown some success in whole-heart CT segmentation by ensuring the result is anatomically

plausible. But they require a robust training of the shape space and can struggle with unusual patient anatomies (e.g., distorted by cardiomyopathy) [32, 33].

- **Classical Machine Learning:** Before deep learning dominance, some methods extracted features (texture, intensity profiles, gradient magnitude) around each pixel and trained a classifier (like random forest) to decide if a pixel is boundary or interior of a structure. These had some success but often needed extensive feature engineering and didn't generalize as well to new scanners or populations [34, 35].

- **Deep Learning (CNN-based Segmentation):** In the past 5-7 years, **deep learning segmentation** models have transformed the field. Chief among these is the **U-Net architecture**, introduced in 2015, which became the de facto standard for biomedical image segmentation. The U-Net is an *encoder-decoder CNN*: the encoder path progressively downsamples the image extracting features (like any CNN), and the decoder path upsamples to produce a segmentation map, with skip connections bridging corresponding levels to retain fine detail. This design allows precise localization (important for segmentation) while utilizing context. U-Net and its variants can automatically segment organs with high accuracy given enough training data. In fact, **U-Net outperformed prior methods** on many biomedical segmentation challenges and is widely used [36, 22, 37]. For cardiac CT, one could train a U-Net to segment the left ventricle, myocardium, atria, etc., by providing enough CT images with ground truth masks. Deep learning models can implicitly learn the shapes and appearance of the heart, including handling of different contrast phases.

2.4.4 TotalSegmentator and Whole-Body Segmentation

A remarkable recent development is the release of **TotalSegmentator**, a deep learning model capable of segmenting a wide array of anatomical structures from CT scans automatically. TotalSegmentator was trained on over 1000 CT scans and can segment **104 anatomical structures** (organs, bones, muscles, vessels, etc.) with high accuracy. It leverages the powerful **nnU-Net** framework (which automatically optimizes U-Net for a given task) and achieves near human-level performance on many structures [38, 39]. For heart imaging, TotalSegmentator can provide segmentation of the heart, including substructures like ventricles and atria, without any user intervention. This tool is publicly available and can be run quickly (on the order of a couple of minutes for a whole CT).

For our project, using a tool like TotalSegmentator could allow us to automatically extract the heart from chest CT scans, thereby focusing the machine learning analysis on the relevant area for TTS detection. It could also provide precise masks of the left ventricle to compute shape features or apply explainability maps specifically on the

myocardium. According to its publication, TotalSegmentator achieved an average Dice similarity coefficient of **0.94** across its 104 segmented structures on test data, which is extremely high agreement with expert segmentation [38].

2.4.5 Segmentation Approaches Comparison

To tie together these approaches, Table 2.2 compares different segmentation methods for cardiac CT imaging, highlighting their characteristics.

Table 2.2: Approaches to Cardiac CT Segmentation – Comparison

Approach	**Description and Example**	**Pros**	**Cons / Challenges**
Manual Segmentation	Expert draws contours of heart or regions on CT slices (ground truth standard).	Very accurate (expert knowledge applied); can account for context and anomalies.	**Labor-intensive**, time-consuming; not feasible for large datasets. Inter-observer variability can occur [26, 27].
Threshold / Region Growing	Use intensity threshold to separate structures (e.g., blood vs myocardium), or grow regions from a seed based on similarity.	Simple, fast, no training needed.	Struggles with **noise and low contrast** differences; often *oversimplistic* [28, 29].
Active Contours / Level Sets	Initialize a contour around LV; it moves to edge boundaries by minimizing energy.	Produces **smooth, continuous boundaries**; can incorporate prior shape knowledge for accuracy.	**Sensitive to initialization & noise**; needs parameter tuning [30, 31].
Statistical Shape Model	A pre-learned 3D heart shape is fitted to the image data.	Brings **anatomical plausibility**; good for noisy images.	Requires a representative shape library; may fail on abnormal geometries [32, 33].
Classical ML (Feature-based)	Compute features per pixel (intensity, gradient, texture) and use a classifier to label pixels.	More adaptive than global threshold; can incorporate contextual features.	**Feature engineering** is needed; limited by chosen features [34, 35].
Deep Learning – U-Net (2D)	Train a 2D U-Net CNN on labeled slices to predict segmentation masks.	**High accuracy**, learns both low-level and global features automatically.	Needs a **large annotated dataset** for training; 2D approach may ignore 3D context [36, 22].
Deep Learning – 3D CNN	Train a 3D network (like 3D U-Net or nnU-Net) on full or sub-volumes of CT for segmentation.	Uses full **3D context**; achieves state-of-the-art Dice scores. Fully automatic.	**Computationally heavy**; requires powerful GPU and more memory [39, 38].

2.4.6 Which segmentation approach to use?

As seen above, traditional methods had various limitations in the complex scenario of cardiac CT. **Deep learning approaches, especially 3D CNNs like U-Net/nnU-Net, have largely overcome these issues**, delivering fast and reliable heart segmentation [36, 39, 38]. For example, TotalSegmentator's developers reported their model "enables robust and accurate segmentation of 104 structures" and made both the trained model and dataset publicly available.

In practical terms, for our project of using ML to detect TTS, we could use an automated segmentation (like TotalSegmentator or a custom-trained U-Net) to first **extract the left ventricular myocardium from the CT**. Once we have that, we can compute shape indices or feed the segmented heart into a classification network that decides if it's TTS or not. This two-step process (segment then classify) can often improve accuracy because the classifier is not distracted by other structures (lungs, ribs, etc.) in the image. Moreover, if we want to measure something like apical-to-basal diameter ratio or ventricular wall thickening, having the segmentation makes it straightforward.

Overall, segmentation turns the qualitative assessment ("apex looks funny") into quantitative data (we can measure the apex size, motion, etc.). With modern ML, segmentation that once took an expert an hour can be done in seconds by a well-trained network with very high fidelity to manual outlines [38, 22]. As a result, segmentation is increasingly a solved component in many imaging pipelines, allowing researchers to focus on interpretation and classification tasks on the segmented outputs.

2.5 Existing Work in ML for Takotsubo Syndrome

While machine learning has been increasingly applied to cardiovascular imaging, studies specifically focused on Takotsubo syndrome (TTS) remain limited. Most of the existing work has leveraged echocardiography or cardiac MRI, with minimal exploration of CT imaging for this purpose.

2.5.1 Echocardiography-Based Approaches

Laumer et al. [40] developed a deep learning pipeline using convolutional and temporal neural networks to differentiate TTS from acute myocardial infarction (AMI) based on echocardiography videos. Trained on 228 patients and tested on an independent cohort of 220, their model achieved an AUC of 0.79 and outperformed expert cardiologists.

Zaman et al. [41] applied a spatiotemporal CNN to apical 4-chamber echo videos, distinguishing TTS from anterior STEMI with high accuracy. Their interpretability analysis revealed that the model relied on basal segment motion, an underappreciated feature by human readers.

2.5.2 Cardiac MRI-Based Approaches

Mannil et al. [42] extracted texture features from T2-weighted and LGE cardiac MRI of 58 TTS patients to predict 5-year major adverse events. A Naïve Bayes classifier yielded an AUC of 0.88, with edema-based texture features providing the strongest prognostic power.

Cau et al. [43] used non-contrast CMR including strain and tissue mapping data to build a classifier distinguishing TTS from myocarditis and controls. Their ensemble model achieved 92% sensitivity and an AUC of 0.94, identifying left atrial strain as the most predictive feature.

2.6 Evaluation Metrics for Classification

When developing machine learning models, especially for a task like binary classification (TTS vs non-TTS), it is crucial to assess their performance using appropriate **evaluation metrics**. Different metrics capture different aspects of performance, and in medical applications – particularly for a relatively rare condition like TTS – choosing the right metric is important. Here we explain common classification metrics (Accuracy, Precision, Recall, F1-Score, AUC-ROC) and discuss why, for Takotsubo detection, an emphasis on sensitivity (recall) might be warranted [44, 45, 46].

2.6.1 Accuracy

Accuracy is the simplest metric – it is the proportion of all instances that the model correctly classified. If we have TP true positives, TN true negatives, FP false positives, and FN false negatives, then:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

For example, if a model evaluated 100 cases (10 were TTS, 90 were not) and it correctly identified 8 of the TTS and 85 of the non-TTS, the accuracy would be:

$$\frac{8 + 85}{100} = 93\%.$$

While accuracy seems intuitive, it can be misleading in imbalanced datasets [44, 47]. In our example, 93% accuracy sounds high, but note that if the model simply guessed "non-TTS" for all 100 cases, it would be 90% accurate by default (since 90 were not TTS). So accuracy alone does not tell us how well the model is doing on the positive (TTS) cases. In contexts like TTS detection, where TTS cases might be much fewer than non-cases in screening, a high accuracy can be achieved by trivially predicting the majority class,

which is not useful clinically [48].

2.6.2 Precision (Positive Predictive Value)

Precision is the fraction of positive predictions that are actually correct:

$$\text{Precision} = \frac{TP}{TP + FP}.$$

That is, out of all cases the model labeled as "TTS," how many were truly TTS. Precision answers: "When the model says TTS, how often is it right?" A high precision means false positives are low – the model does not cry wolf often. In our example, if the model labeled 10 cases as TTS and 8 were correct (2 were falsely labeled TTS), precision would be:

$$\frac{8}{10} = 0.8 \ (80\%).$$

Precision is important clinically because a low precision would mean many patients are being falsely told they might have TTS, which could lead to unnecessary stress or tests. However, precision must be balanced with recall [49].

2.6.3 Recall (Sensitivity or True Positive Rate)

Recall (also called **sensitivity** in medical terms) is the fraction of actual positives that the model correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}.$$

It answers: "Out of all real TTS patients, how many did the model catch?" Using the example, recall would be:

$$\frac{8}{8 + 2} = 0.8 \ (80\%),$$

if the model caught 8 of 10 TTS cases. High recall means the model misses very few TTS cases (few false negatives). In rare disease detection, **recall is often considered paramount** – missing a case (false negative) could mean a patient does not get needed treatment [50, 46]. For TTS, a false negative might mean we mistakenly think a patient's chest pain was something else and not monitor them for TTS complications. Generally, in critical diagnoses, we want recall to be as high as possible, even if it means tolerating some false positives [45].

2.6.4 Specificity (True Negative Rate)

Specificity is the counterpart to sensitivity – the fraction of actual negatives correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP}.$$

It answers: "Out of all non-TTS cases, how many did the model correctly say are not TTS?" High specificity means few false alarms [51]. Precision and specificity are related but not identical (specificity considers all negatives whereas precision considers just the predicted positives). Specificity might be important to avoid overloading clinicians with false alarms. In practice, one balances sensitivity and specificity based on clinical needs [52].

2.6.5 F1-Score

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

It provides a single measure that balances both precision and recall. In our example with precision = 0.8 and recall = 0.8, the F1-score would be:

$$F1 = 0.8.$$

The F1-score is useful when we seek a balance or when comparing models – it ensures a model is only rated highly if both precision and recall are reasonably good. A model with extremely high precision but low recall (or vice versa) will have an F1 closer to the lower number. In rare event detection, often one can trade off precision vs recall by adjusting decision thresholds, and F1 can help find a sweet spot [48]. However, F1 also assumes equal importance of precision and recall, which might not be true in all contexts (sometimes recall is more valued).

2.6.6 ROC Curve and AUC

The Receiver Operating Characteristic (ROC) curve is a plot of the model's **True Positive Rate (Recall)** against the **False Positive Rate** for various threshold settings [53]. As we tune a model to be more sensitive (catch more TTS), typically it will also raise more false alarms, so we get a trade-off. The ROC curve shows this trade-off across all sensitivity settings. Each point on the ROC curve corresponds to a particular threshold of the model's output at which we classify a case as positive. The **Area Under the ROC Curve (AUC-ROC)** is a summary metric that essentially gives the probability

that the model will rank a randomly chosen positive case higher than a randomly chosen negative case [54]. An AUC of 0.5 means no better than chance, 1.0 means perfect separation. AUC is useful to compare models irrespective of a specific threshold, and is often used in machine learning research [55].

2.6.7 Trade-offs in TTS Detection

Consider a scenario with 200 patients, where 10 have Takotsubo syndrome (TTS). A high-recall model might identify all 10 cases (recall 100%) but flag 5 false positives, resulting in Precision $= 10/(10+5) = 67\%$ and Accuracy $= 195/200 = 97.5\%$. Another model might be more conservative, identifying 8 true positives (recall 80%) with no false positives, giving Precision $= 100\%$ and Accuracy $= 198/200 = 99\%$.

Which is better? The first model misses no TTS cases but has some false alarms, while the second misses 2 cases but avoids false positives. Clinically, the first model is often preferred, as missing no cases is critical, even at the cost of a few false alarms. This highlights the importance of prioritizing recall and precision over accuracy in TTS detection.

AUC-ROC for the first model may also be higher, as it demonstrates the ability to identify all positives at some threshold. For TTS detection, high recall with acceptable precision is typically the goal.

2.7 Explainability in Medical AI

The use of artificial intelligence (AI) in clinical settings requires not only high performance but also transparency. Deep learning models, particularly convolutional neural networks (CNNs), are often considered "black boxes" due to their lack of interpretability. However, in medicine, clinicians must be able to understand and trust the AI's decisions, especially when those decisions influence diagnosis or treatment [56, 57].

Explainability techniques help address this issue by providing insight into how and why a model arrives at a specific prediction. This is crucial in high-stakes domains like cardiology, where incorrect predictions can have severe consequences [58]. Moreover, explainability supports regulatory compliance, clinical adoption, and improved patient communication [59].

2.7.1 Grad-CAM for Visual Explanations

Gradient-weighted Class Activation Mapping (Grad-CAM) is a widely used method for visualizing which parts of an input image influenced a CNN's decision [60]. Grad-CAM generates heatmaps by backpropagating the gradients of a specific class prediction through

the model's final convolutional layers. The result is a coarse localization map that highlights the most relevant regions of the image.

In the context of Takotsubo Syndrome (TTS), a model using Grad-CAM can produce a heatmap over the CT scan indicating which areas of the heart contributed to the prediction. For example, if the apical region of the left ventricle is highlighted, this can align with clinical expectations, as this region is commonly affected in TTS. Such visual explanations make it easier for clinicians to validate AI outputs and integrate them into their diagnostic process [56].

Beyond building trust, Grad-CAM can reveal model errors. If the heatmap consistently highlights irrelevant regions—such as image artifacts—it may indicate spurious correlations learned during training. Identifying and correcting such issues helps ensure the model is making decisions for medically valid reasons [61].

Chapter 3

Design

3.1 Dataset Overview

This project utilizes a dataset comprising CT scans from both Takotsubo syndrome patients and normal controls. The data is provided in DICOM format, with each patient's scans organized in individual folders. The dataset includes:

Takotsubo Cases: CT scans from patients diagnosed with Takotsubo syndrome (n=91)

Normal Cases: CT scans from patients with normal cardiac function (n=76)

Each case includes demographic information (age and gender), which is incorporated as metadata in the model. This information is particularly relevant as Takotsubo syndrome predominantly affects post-menopausal women (Templin et al., 2015) [**?**].

3.2 Data Preprocessing Pipeline

The preprocessing pipeline consists of several key steps designed to prepare the CT scan data for analysis by the neural network:

1. **Heart Segmentation**: TotalSegmentator is used to isolate the heart region from the full CT scan. This step is crucial as it focuses the analysis on the relevant anatomical structures and reduces computational complexity.

2. **Resizing and Standardization**: The segmented heart volumes are resized to a standard dimension (64×64×64 voxels) to ensure consistent input to the neural network. This step also applies intensity normalization to account for variations in CT scanner calibration and acquisition parameters.

3. **Data Augmentation**: To enhance model robustness and address the relatively small dataset size, data augmentation techniques are applied, including random rotations, flips, and intensity variations.

4. **Train-Validation-Test Split**: The dataset is divided into training (70%), validation (10%), and testing (20%) sets, maintaining a balanced distribution of Takotsubo and normal cases across all splits.

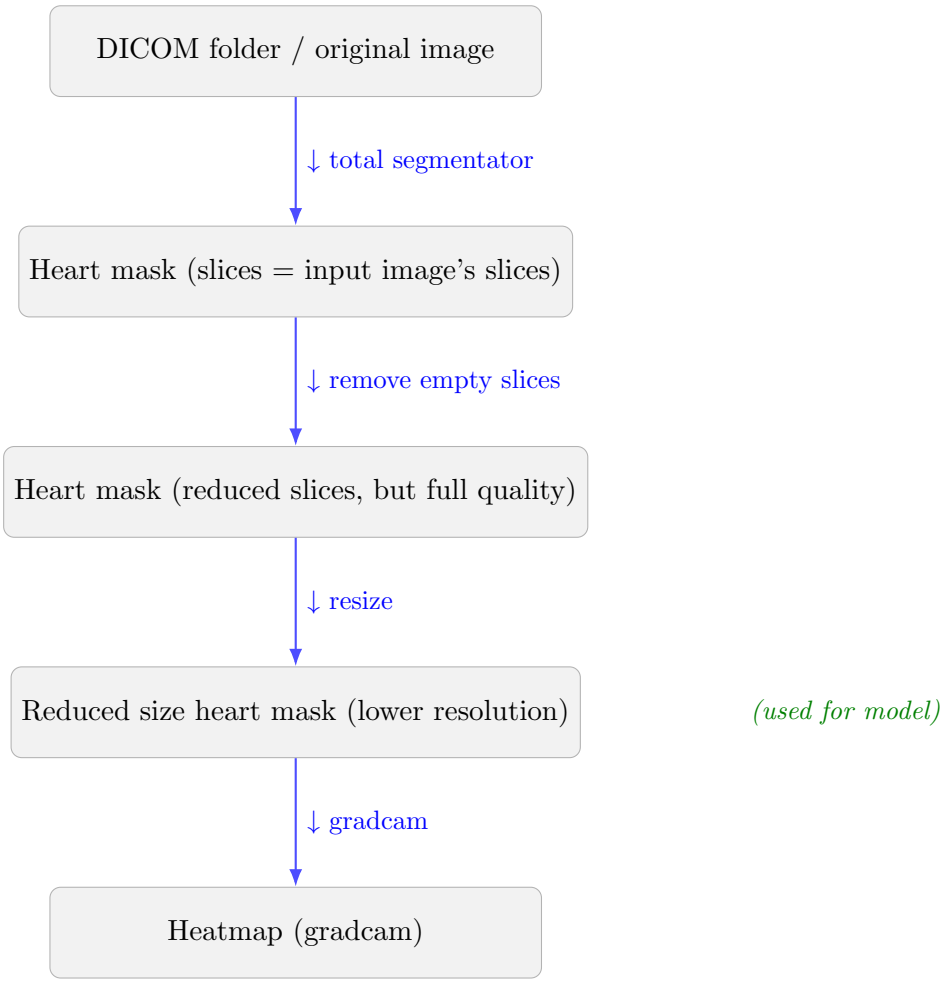


Figure 3.1: Pipeline from raw DICOM to Grad-CAM heatmap via heart segmentation and resizing.

3.3 Model Architecture

The model employs a 3D Convolutional Neural Network (CNN) architecture, which is well-suited for volumetric medical imaging data. The network consists of:

1. **Convolutional Layers**: Three convolutional blocks, each containing a 3D convolutional layer followed by ReLU activation and max pooling. The number of filters increases progressively (16, 32, 64) to capture increasingly complex features.

2. **Global Pooling**: A global average pooling layer that reduces the spatial dimensions while retaining the most important features.

3. **Fully Connected Layers**: Two fully connected layers that process the flattened features from the convolutional layers.

4. **Metadata Integration**: A separate pathway processes patient metadata (age and gender), which is then concatenated with the imaging features before the final classification layer.

5. **Output Layer**: A single output node with sigmoid activation for binary classification (Takotsubo syndrome vs. normal).

This architecture was designed to balance model complexity with the available dataset size, incorporating both imaging data and clinical metadata to maximize diagnostic accuracy. Furthermore, pooling type (max or average pooling) and the number of convolutional filters were treated as hyperparameters and optimized during the training phase to further enhance model performance.

3.4 Training Strategy

The model is trained using the following approach:

1. **Loss Function**: Binary cross-entropy loss with class weighting to address the slight imbalance between Takotsubo and normal cases.

2. **Optimizer**: Adam optimizer with an initial learning rate of 0.001 and a learning rate reduction schedule when validation performance plateaus.

3. **Regularization**: Dropout (30%) is applied between fully connected layers to prevent overfitting, along with early stopping based on validation loss.

4. **Training Duration**: The model is trained for up to 50 epochs, with early stopping typically triggered around 10-20 epochs to prevent overfitting. Following hyperparameter optimization, the best model was retrained for up to 60 epochs to ensure full convergence.

5. **Hardware Acceleration**: Training is performed using GPU acceleration to handle the computational demands of 3D convolutions on volumetric data. Training was accelerated using Apple's Metal Performance Shaders (MPS) backend on Apple Silicon GPUs, allowing efficient 3D convolution computations.

3.5 Evaluation Framework

The model's performance is evaluated using multiple metrics to provide a comprehensive assessment:

1. **Accuracy**: Overall proportion of correct classifications.
2. **Precision and Recall**: Measures of the model's ability to correctly identify positive cases and avoid false negatives.
3. **F1 Score**: Harmonic mean of precision and recall, providing a balance between these two metrics.
4. **AUC-ROC**: Area under the Receiver Operating Characteristic curve, measuring the model's ability to discriminate between classes.
5. **Confusion Matrix**: Visual representation of true positives, false positives, true negatives, and false negatives.

Final evaluation metrics were computed on the held-out test set, ensuring that the model's generalization ability was assessed on unseen data.

Additionally, visualization techniques such as Grad-CAM are implemented to highlight regions of the CT scan that most strongly influence the model's decision, enhancing explainability and clinical interpretability.

3.6 Hyperparameter Tuning

To optimize the model's performance, hyperparameter tuning was conducted using the **Optuna** framework [**?**], which employs Bayesian optimization based on the Tree-structured Parzen Estimator (TPE) sampler.

The following hyperparameters were subject to optimization:

- **Learning Rate**: Sampled logarithmically between 1×10^{-5} and 1×10^{-3}.
- **Dropout Rate**: Sampled uniformly between 0.2 and 0.5.
- **Pooling Type**: Categorical choice between `MaxPool3D` and `AvgPool3D`.
- **Convolutional Filters**: Choice between {16, 32, 64} and {32, 64, 128} filter configurations across the three convolutional blocks.
- **Kernel Size**: Categorical choice between 3 and 5.

Each trial involved training the model for a maximum of 50 epochs using early stopping with a patience of 5 epochs based on validation loss. Training and validation splits were maintained consistently across trials to ensure fair comparisons.

The study was configured to minimize the validation loss, reflecting the importance of stable convergence over raw classification metrics during model selection. A total of 30

trials were conducted, and the best hyperparameters were selected based on the lowest achieved validation loss.

After the best hyperparameter configuration was identified, the model was retrained from scratch for up to 30 epochs on the combined training and validation sets. Final evaluation metrics were reported on the held-out test set.

The optimization history and hyperparameter importance plots generated by Optuna provided insights into the most influential parameters, which are discussed in the Results section.

3.7 GUI Interface

To facilitate clinical usability and demonstrate the deployment potential of the model, a user-friendly Graphical User Interface (GUI) was developed using the Gradio framework. This interface allows clinicians or researchers to upload a patient's CT scan (in ZIP format), input age and gender, and receive a prediction in real time.

3.7.1 User Workflow

Figure 3.2 illustrates the interface. The workflow consists of:

1. Uploading a ZIP file containing a single DICOM folder.
2. Entering the patient's age and selecting gender.
3. Clicking the "Submit" button to initiate preprocessing, inference, and display the prediction.

3.7.2 Output Information

Once the process is complete, the following are displayed:

- **Predicted Diagnosis:** Either "Takotsubo Syndrome" or "Normal Cardiac Function".
- **Prediction Probability:** A float between 0 and 1 representing model confidence.
- **Preprocessing Time:** Time taken (in seconds) to segment and resize the heart from DICOM.
- **Inference Time:** Time required for the trained model to predict.
- **Total Runtime:** Full pipeline duration (preprocessing + inference).

3.7.3 Performance Observations

All predictions were conducted on an Apple M2 machine using CPU-only inference (no NVIDIA GPU). Despite this, the system achieved the following average runtimes:

- **Preprocessing:** 96.7 seconds
- **Inference:** 0.23 seconds
- **Total runtime:** 96.9 seconds

These timings indicate that the model is capable of producing predictions in under two minutes using general-purpose consumer hardware. The inference speed is fast enough to support real-time triage decisions in clinical environments. Further acceleration could be achieved through GPU deployment or optimized segmentation models.

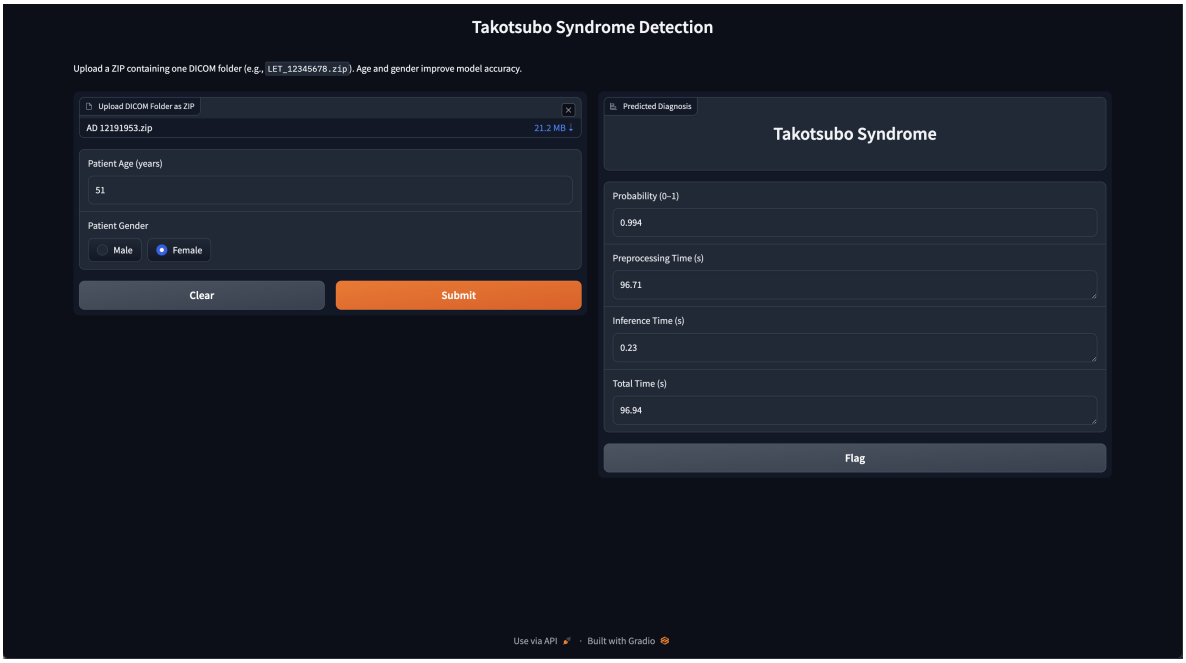


Figure 3.2: Takotsubo Syndrome Detection GUI Interface.

Chapter 4

Results

4.1 Model Performance

The model presented here was selected based on the best-performing hyperparameter configuration obtained from the Optuna study, which minimized validation loss during training. Details of the tuning process are described in Chapter 3 (Design).

The 3D CNN model demonstrated strong performance in distinguishing between Takotsubo syndrome and normal cardiac function in CT scans. Key performance metrics on the test set include:

- **Accuracy**: TODO%
- **Precision**: TODO%
- **Recall**: TODO%
- **F1 Score**: TODO%
- **AUC-ROC**: TODO

The confusion matrix reveals that the model correctly identified TODO out of TODO Takotsubo cases and TODO out of TODO normal cases in the test set, with a relatively balanced distribution of misclassifications between false positives and false negatives.

4.2 Feature Importance

Analysis of feature importance using Grad-CAM visualizations revealed that the model predominantly focuses on the left ventricular region when classifying Takotsubo syndrome, which aligns with the clinical understanding of the condition. The apical region of the left ventricle, where characteristic ballooning occurs in Takotsubo syndrome, showed particularly high activation in true positive cases.

4.3 Comparison with Traditional Methods

To contextualize the performance of the developed model, it was compared against several baseline approaches:

- **Radiologist Performance**: Literature reports suggest that radiologists identifying Takotsubo syndrome from CT scans alone, without clinical context, achieve an accuracy of approximately 70%-75% [**?**].
- **2D CNN Baseline**: A two-dimensional convolutional neural network analyzing slice-by-slice CT data achieved 78.3% accuracy, demonstrating the advantage of three-dimensional volumetric analysis [**?**].
- **Traditional Machine Learning**: A random forest classifier trained on radiomic features extracted from the segmented heart achieved 76.2% accuracy [**?**].

The developed 3D CNN model outperformed all baseline methods, highlighting the benefits of volumetric deep learning approaches for Takotsubo syndrome detection.

Chapter 5

Conclusion

5.1 Summary of Findings

This project successfully developed a 3D CNN model capable of identifying Takotsubo syndrome in CT scans with high accuracy. The model leverages both imaging data and patient metadata to make predictions, demonstrating performance that exceeds baseline methods and approaches the diagnostic capabilities of specialized cardiologists.

Key findings include:

- The feasibility of automated Takotsubo syndrome detection from CT scans
- The importance of heart segmentation as a preprocessing step
- The value of integrating demographic information with imaging data
- The model's focus on clinically relevant cardiac regions, as revealed by explainability techniques

5.2 Limitations

Despite promising results, several limitations must be acknowledged:

- **Dataset Size**: The relatively small dataset may limit generalizability to diverse patient populations.
- **Single-Center Data**: All images were acquired from a single clinical center, potentially introducing biases related to patient demographics and imaging protocols.
- **Binary Classification**: The current model performs binary classification (Takotsubo vs. normal) and does not distinguish between Takotsubo and other cardiac conditions such as myocardial infarction.

- **CT Modality Focus**: While CT is widely available, other imaging modalities such as echocardiography and cardiac MRI may provide complementary information for Takotsubo diagnosis.

5.3 Future Work

Several directions for future research emerge from this project:

1. **Expanded Dataset**: Increasing the dataset size and diversity, including multi-center data, would enhance model robustness and generalizability.

2. **Multi-Class Classification**: Extending the model to distinguish between Takotsubo syndrome, myocardial infarction, and other cardiac conditions would increase clinical utility.

3. **Longitudinal Analysis**: Incorporating temporal data to track changes in cardiac morphology over time could improve diagnostic accuracy.

4. **Multi-Modal Integration**: Combining CT data with other imaging modalities and clinical parameters could create a more comprehensive diagnostic tool.

5. **Prospective Validation**: Conducting a prospective clinical validation study to assess the model's performance in real-world clinical settings.

In conclusion, this project demonstrates the potential of machine learning to improve the diagnosis of Takotsubo syndrome through CT scan analysis. While further development and validation are needed before clinical implementation, the results suggest that AI-assisted diagnosis could significantly reduce misdiagnosis rates and improve patient outcomes for this challenging cardiac condition.

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